DSA0601 Data Handling and Visualization

Slot A

CO3_AT1_Assessment Tool: EXPERIMENT

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Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each experiment, use an appropriate visualization, justify the design, interpret the result, and identify potential visualization bias. R/ggplot2 or an equivalent visualization tool may be used.

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318 **Tasks:**

1. Construct an ECDF and a normal Q-Q plot.
2. Use both plots to evaluate the normality claim.
3. Identify evidence of skewness and/or extreme observations.
4. Recommend percentile-based limits or mean ± standard-deviation limits, with justification.

Key competency: ECDF, Q-Q plot, skewness, outliers, distribution interpretation

Aim

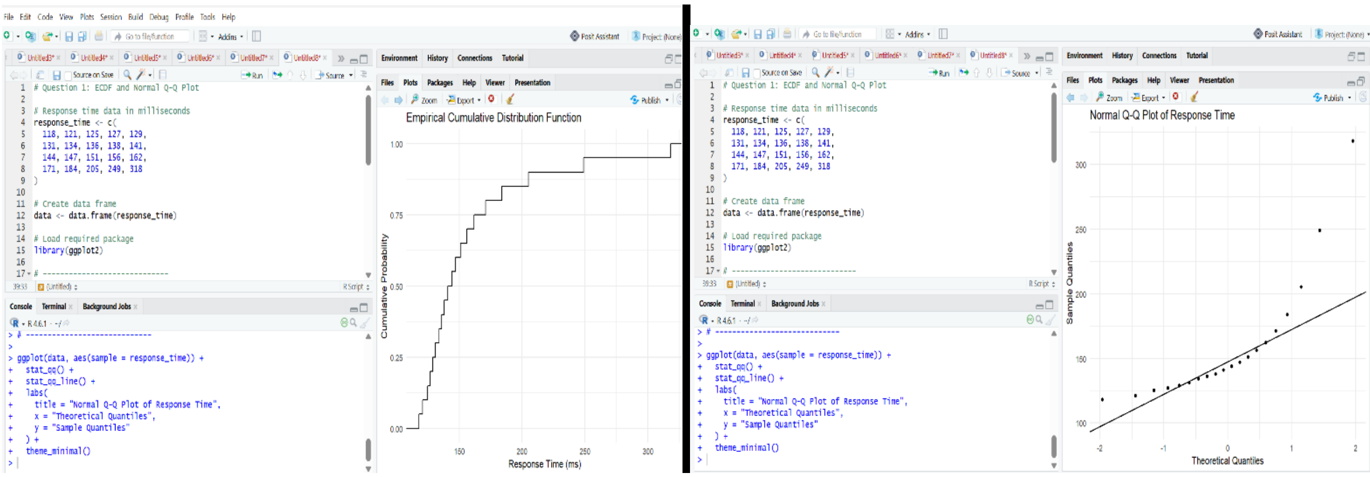
To analyse the response-time distribution of an automated fault-detection system using an ECDF plot and Normal Q-Q plot, and determine whether the data is approximately normally distributed.

R Program

```
response_time <- c(
  118, 121, 125, 127, 129,
  131, 134, 136, 138, 141,
  144, 147, 151, 156, 162,
  171, 184, 205, 249, 318
)
data <- data.frame(response_time)
library(ggplot2)
ggplot(data, aes(x = response_time)) +
  stat_ecdf(geom = "step") +
  labs(
    title = "Empirical Cumulative Distribution Function",
    x = "Response Time (ms)",
    y = "Cumulative Probability"
  ) +
  theme_minimal()
ggplot(data, aes(sample = response_time)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Normal Q-Q Plot of Response Time",
    x = "Theoretical Quantiles",
```

```
    y = "Sample Quantiles"
  ) +
  theme_minimal()
```

Output:



Interpretation

The response-time data is not perfectly normally distributed.

The ECDF shows that most response times are relatively low, while a few observations have considerably larger values. The values 249 ms and 318 ms are particularly extreme.

The Q-Q plot is expected to show deviations from the reference line, especially in the upper tail. Therefore, the normality assumption is questionable.

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

Production Line	**Cycle Times (minutes)**
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

1. Create a visualization comparing all four distributions.
2. Construct bean plots or an equivalent distribution visualization along the vertical axis.
3. Determine which line has greatest variability and which has the most concentrated distribution. 4. Explain why comparing only means is insufficient.

Key competency: Multiple distributions, vertical-axis distributions, bean plots, variability

Aim

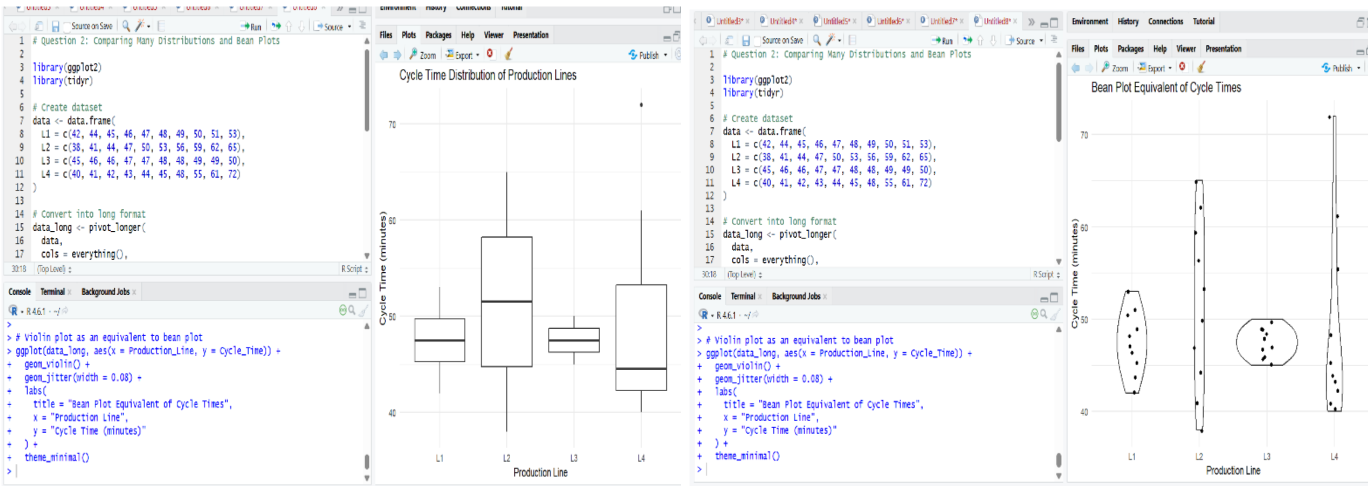
To compare the cycle-time distributions of four production lines and identify differences in variability, concentration, and distribution shape using appropriate distribution visualizations.

Program

```
library(ggplot2)
library(tidyr)
data <- data.frame(
  L1 = c(42, 44, 45, 46, 47, 48, 49, 50, 51, 53),
  L2 = c(38, 41, 44, 47, 50, 53, 56, 59, 62, 65),
  L3 = c(45, 46, 46, 47, 47, 48, 48, 49, 49, 50),
  L4 = c(40, 41, 42, 43, 44, 45, 48, 55, 61, 72)
)
data_long <- pivot_longer(
  data,
  cols = everything(),
  names_to = "Production_Line",
  values_to = "Cycle_Time"
)
ggplot(data_long, aes(x = Production_Line, y = Cycle_Time)) +
  geom_boxplot() +
  labs(
    title = "Cycle Time Distribution of Production Lines",
    x = "Production Line",
    y = "Cycle Time (minutes)"
  ) +
  theme_minimal()
ggplot(data_long, aes(x = Production_Line, y = Cycle_Time)) +
  geom_violin() +
  geom_jitter(width = 0.08) +
  labs(
    title = "Bean Plot Equivalent of Cycle Times",
    x = "Production Line",
    y = "Cycle Time (minutes)"
```

```
  ) +
  theme_minimal()
```

Output:



Interpretation

The four production lines show different levels of variability and distribution shape.

L4 has the greatest variability, with cycle times ranging from 40 to 72 minutes. L3 has the most concentrated distribution, with most values between 45 and 50 minutes.

The distribution plot also shows that L4 has some unusually high values, while L3 is more consistent. Therefore, comparing only the mean is insufficient because it does not show the spread, concentration, or extreme observations.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30

Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.
2. Explain which chart better supports comparison of individual proportions and why.
3. Explain why separate pie charts may be less effective for comparing departments.
4. Identify one design choice that could make small proportional differences appear larger.

Key competency: Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions

Aim

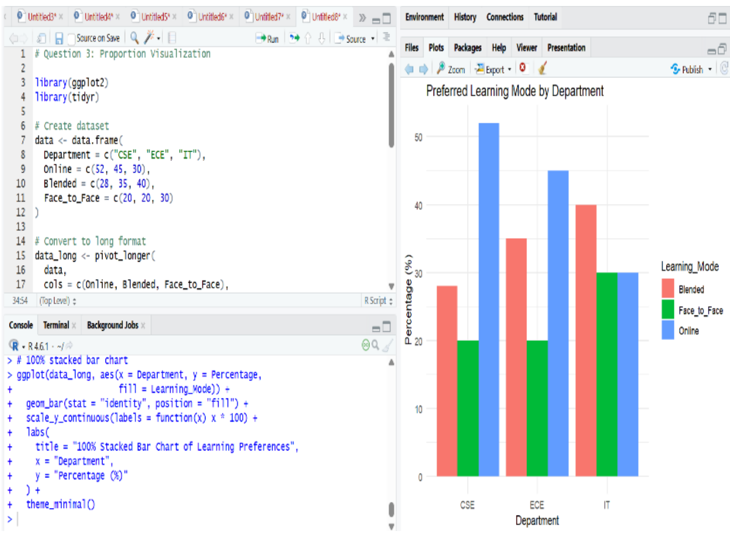
To compare students' preferred learning modes across different departments using side-by-side bar charts and 100% stacked bar charts, while avoiding misleading proportional comparisons.

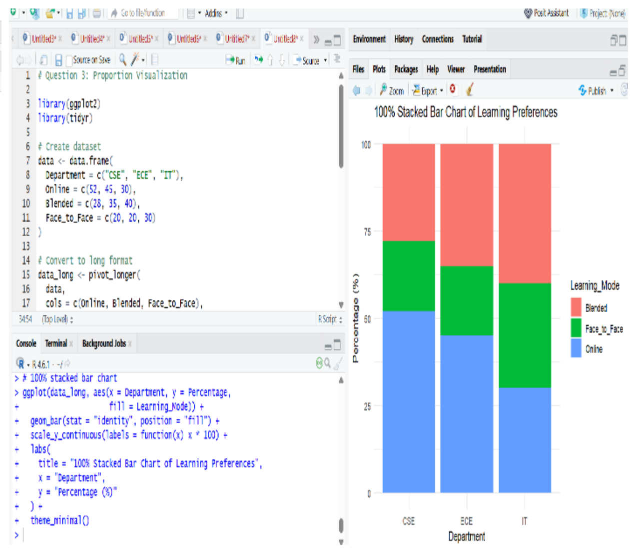
Program

```
library(ggplot2)
library(tidyr)
data <- data.frame(
  Department = c("CSE", "ECE", "IT"),
  Online = c(52, 45, 30),
  Blended = c(28, 35, 40),
  Face_to_Face = c(20, 20, 30)
)
data_long <- pivot_longer(
  data,
  cols = c(Online, Blended, Face_to_Face),
  names_to = "Learning_Mode",
  values_to = "Percentage"
)
ggplot(data_long, aes(x = Department, y = Percentage,
                      fill = Learning_Mode)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Preferred Learning Mode by Department",
    x = "Department",
    y = "Percentage (%)"
  ) +
  theme_minimal()
ggplot(data_long, aes(x = Department, y = Percentage,
                      fill = Learning_Mode)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_y_continuous(labels = function(x) x * 100) +
  labs(
    title = "100% Stacked Bar Chart of Learning Preferences",
```

```
    x = "Department",
    y = "Percentage (%)"
  ) +
  theme_minimal()
```

Output





Interpretation

The side-by-side bar chart makes it easier to compare individual learning-mode percentages across departments because the bars share a common baseline.

The 100% stacked bar chart clearly shows the overall composition of learning preferences within each department, but comparing individual segments is more difficult when they do not start from the same baseline.

Separate pie charts are less effective because comparing angles and areas across multiple pies is harder than comparing aligned bars.

A misleading design choice would be starting the bar chart y-axis above zero or using a very narrow y-axis range, which can make small differences between percentages appear much larger than they actually are.

Question 4 – Nested Proportions using Tree map and Sunburst

Scenario: An e-commerce company analyses annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	Subcategory	Revenue (₹ lakh)

Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks:

1. Construct a tree map showing category and subcategory revenue.
2. Construct a sunburst chart representing the same hierarchy.
3. Determine the largest category and calculate its percentage of total revenue.
4. Compare tree map and sunburst for identifying nested proportions.
5. State one situation in which area encoding could mislead interpretation.

Key competency: Tree maps, sunburst charts, nested proportions, area encoding

Aim

To visualize annual revenue using tree map and sunburst charts to compare category contributions and their subcategory composition.

Program

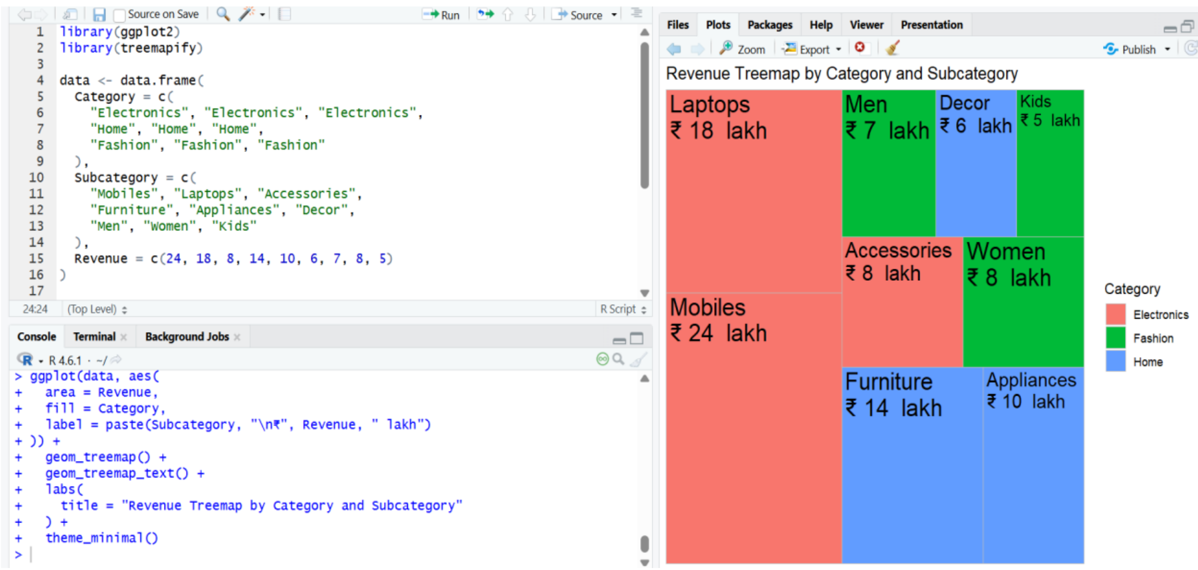
```
library(ggplot2)
library(treemapify)
library(plotly)
data <- data.frame(
  Category = c(
    "Electronics", "Electronics", "Electronics",
```

```
    "Home", "Home", "Home",
    "Fashion", "Fashion", "Fashion"
  ),
  Subcategory = c(
    "Mobiles", "Laptops", "Accessories",
    "Furniture", "Appliances", "Decor",
    "Men", "Women", "Kids"
  ),
  Revenue = c(24, 18, 8, 14, 10, 6, 7, 8, 5)
)
ggplot(data, aes(
  area = Revenue,
  fill = Category,
  label = paste(Subcategory, "\n₹", Revenue, " lakh")
)) +
  geom_treemap() +
  geom_treemap_text(colour = "white") +
  labs(
    title = "Revenue Treemap by Category and Subcategory"
  ) +
  theme_minimal()
sunburst_data <- data.frame(
  ids = c(
    "Total",
    "Electronics", "Mobiles", "Laptops", "Accessories",
    "Home", "Furniture", "Appliances", "Decor",
    "Fashion", "Men", "Women", "Kids"
  ),
  labels = c(
    "Total",
    "Electronics", "Mobiles", "Laptops", "Accessories",
    "Home", "Furniture", "Appliances", "Decor",
    "Fashion", "Men", "Women", "Kids"
  ),
  parents = c(
    "",
    "Total", "Electronics", "Electronics", "Electronics",
    "Total", "Home", "Home", "Home",
    "Total", "Fashion", "Fashion", "Fashion"
  ),
  values = c(
    100,
    50, 24, 18, 8,
    30, 14, 10, 6,
    20, 7, 8, 5
  )
)

plot_ly(
```

```
  data = sunburst_data,
  ids = ~ids,
  labels = ~labels,
  parents = ~parents,
  values = ~values,
  type = "sunburst"
) %>%
  layout(
    title = "Sunburst Chart of Revenue"
  )
```

Output



Interpretation

The treemap shows that Electronics is the largest category, with revenue of ₹50 lakh, followed by Home with ₹30 lakh and Fashion with ₹20 lakh.

The total revenue represented by the given data is ₹100 lakh (₹1 crore). Therefore, Electronics contributes 50% of the total revenue.

The treemap makes it easy to compare the area of categories and subcategories, while the sunburst clearly shows the hierarchical relationship between categories and their subcategories.

Area encoding can sometimes mislead interpretation because people may find it difficult to accurately compare the sizes of similarly sized areas, especially when many small subcategories are present.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets Scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks:

1. Construct a Sankey diagram showing Origin → Mode → Destination.
2. Construct a parallel-sets representation of the same categorical flows.
3. Identify the three largest flows and calculate their percentages of all commuters.
4. Compare Sankey and parallel sets for flow magnitude and category relationships.
5. Explain one way poor link widths or category ordering could mislead the viewer.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships

Aim

To visualize the flow of commuters from origin to transport mode to destination using a Sankey diagram and identify the major commuter flows.

Program

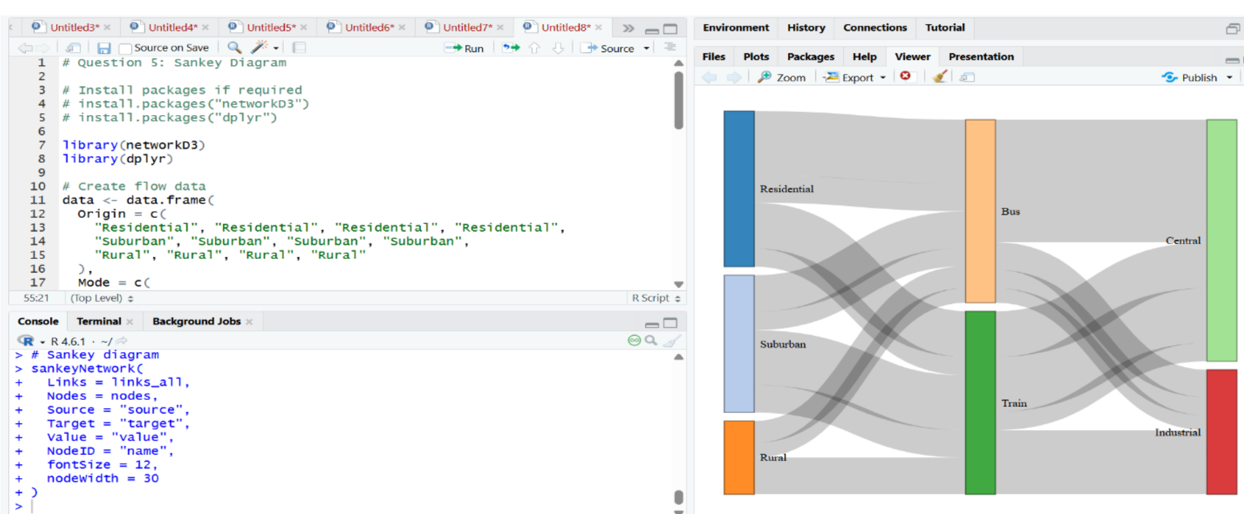
```
library(networkD3)
library(dplyr)
data <- data.frame(
  Origin = c(
    "Residential", "Residential", "Residential", "Residential",
    "Suburban", "Suburban", "Suburban", "Suburban",
    "Rural", "Rural", "Rural", "Rural"
  ),
  Mode = c(
    "Bus", "Bus", "Train", "Train",
    "Bus", "Bus", "Train", "Train",
    "Bus", "Bus", "Train", "Train"
  ),
  Destination = c(
    "Central", "Industrial", "Central", "Industrial",
    "Central", "Industrial", "Central", "Industrial",
    "Central", "Industrial", "Central", "Industrial"
  ),
  Commuters = c(35, 15, 25, 10, 20, 10, 30, 15, 12, 8, 10, 10)
)
nodes <- data.frame(
  name = unique(c(data$Origin, data$Mode, data$Destination))
)
links <- data.frame(
  source = match(data$Origin, nodes$name) - 1,
  target = match(data$Mode, nodes$name) - 1,
  value = data$Commuters
)
links2 <- data.frame(
  source = match(data$Mode, nodes$name) - 1,
  target = match(data$Destination, nodes$name) - 1,
  value = data$Commuters
)
links_all <- rbind(links, links2)
sankeyNetwork(
  Links = links_all,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
```

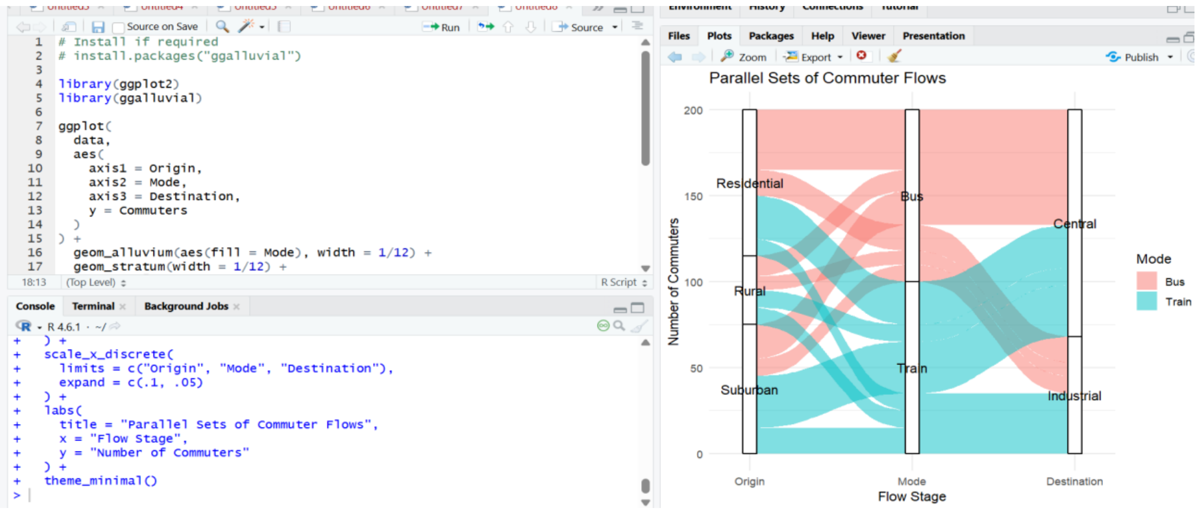
```
  nodeWidth = 30
)
```

Parallel Sets Representation

```
library(ggplot2)
library(ggalluvial)
ggplot(
  data,
  aes(
    axis1 = Origin,
    axis2 = Mode,
    axis3 = Destination,
    y = Commuters
  )
) +
  geom_alluvium(aes(fill = Mode), width = 1/12) +
  geom_stratum(width = 1/12) +
  geom_text(
    stat = "stratum",
    aes(label = after_stat(stratum))
  ) +
  scale_x_discrete(
    limits = c("Origin", "Mode", "Destination"),
    expand = c(.1, .05)
  ) +
  labs(
    title = "Parallel Sets of Commuter Flows",
    x = "Flow Stage",
    y = "Number of Commuters"
  ) +
  theme_minimal()
```

Output





Interpretation

The Sankey diagram clearly shows the flow of commuters from their origin through transport mode to destination.

The three largest individual flows are Residential → Bus → Central (35 commuters), Suburban → Train → Central (30 commuters), and Residential → Train → Central (25 commuters). Since there are 200 commuters, these represent 17.5%, 15%, and 12.5% respectively.

The Sankey diagram is useful for understanding flow magnitude, because wider links represent more commuters. The parallel-sets plot is useful for seeing relationships between categorical variables across the three stages.

Poorly scaled link widths or an unsuitable ordering of categories could make some flows appear more or less important than they actually are.

Suggested Evaluation Rubric – Experiment

Criterion	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.